A Two-Stage Latent Diffusion Transformer for Brain-Metastasis Progression Synthesis: Bounded, Reproducible, and Sampler-Verified

Jinyu He
Zhejiang University of Science and Technology, Hangzhou, China
Code & checkpoints: github.com/fgn2pyh2wn-afk/brain_tumour_predict

1. Motivation

Brain metastases are the commonest intracranial secondary tumours, and management turns on how lesions change between follow-ups, yet repeat imaging is scarce. We present a lightweight two-stage latent diffusion model (LDM) [2] that synthesises T1-Gd progression as a bounded, reproducible 64^3 sequence from one scan: a custom-trained 3D VAE and a Diffusion Transformer (DiT) [3] predict latent noise, and a prescribed growth law replaces uniform step interpolation on a generation grid, not clinical time — small by design, not a biophysically conditioned framework [4]. To keep the generative claims checkable, we add a generation-validation audit (§7) that re-runs every reported number under sampler surrogates and reports which ones are genuinely sampler-attributed.

2. Method

A 3D convolutional autoencoder maps a 64×64×64 T1-Gd volume to a 4×16×16×16 latent; a Diffusion Transformer (dim 64, 2 blocks, 4 heads, patch 2×2×2, 512 tokens; T = 200, linear β $1\times10^{-4}\to0.02$ [1]) predicts latent noise conditioned on a globally pooled latent. Stage 1 picks the starting noise level from the prescribed step span (AdaNI); stage 2 re-noises to $0.15T$ and refines texture. A structure-preserving post-process confines the residual to a lesion-candidate mask and clips it at c = 0.08 with strength s = 0.35; all components are custom-trained.

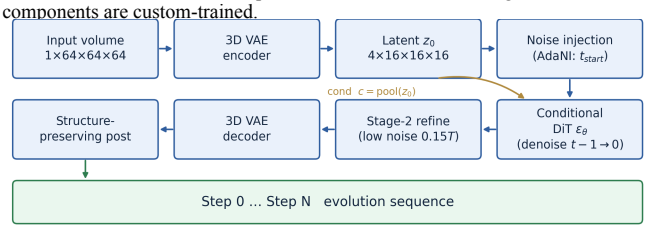


Fig. 1. Two-stage latent-diffusion framework (schematic); every component is custom-trained.

3. Growth-law time weighting

Endpoint weights follow an explicit growth law (exponential / Gompertz / logistic-Fisher-KPP) with an equivalent doubling time T_d instead of uniform step/N interpolation; this is a monotone re-parameterisation of the step axis, not a cohort-calibrated growth model, but one with closed-form structure. Five derived properties are verified on all 51 cases — load–weight affinity (residual 1.02×10^{-15}), a convex-combination identity (1.92×10^{-7}), an adjacent-increment bound, the closed-form maximiser (t^* = 56.30, rel. 4.4×10^{-8}) and boundedness and $\|x(t)-x_0\| = w(t)\cdot\|x_N-x_0\|$ with $|\Delta_{\text{gen}}| \le s\cdot c = 0.028$.

4. Results: structure preservation (Fig. 2)

As trained (before the §8 fix), on N = 100 steps over 51 cases: soft contour IoU 0.9836±0.0047, hard IoU 0.9881±0.0035, texture correlation 0.9808±0.0012, **candidate-region** mean change −0.0219±0.0023 — the region is the heuristic candidate field (24.4% of brain), **not** the lesion; outside/inside ratio 0.633 (0.416–0.863); pre-clip reconstruction PSNR 15.993±1.497 dB.

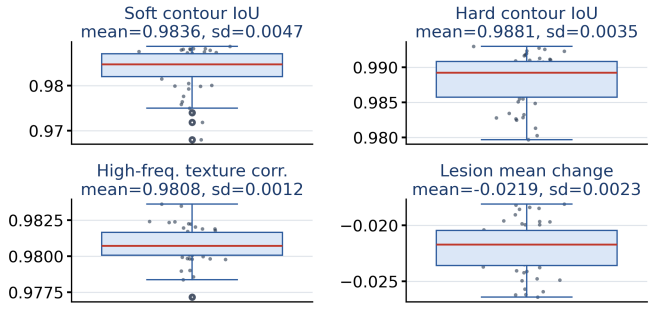


Fig. 2. Progression metrics over the 51-case test set (as-trained weights, i.e. before the §8 fix; N=100 steps; box=IQR, dots=per-case).

5. Expert-lesion cross-check (n = 105; not held-out)

Expert *seg* labels exist only in the training split (the 105 studies it trained on): a cross-check, not held-out performance. The candidate field is 125× the lesion, covering 29.0% of lesion voxels though only 0.37% of it is lesion. Inside expert masks (trilinear / max-pool) the same mean change is −0.0494±0.0237 / −0.0239±0.0090 vs. −0.0214±0.0025: up to 2.3×, and the §7 audit does not move.

6. Qualitative progression (Fig. 3)

Three cases, Step 0→100, through the code's own display chain (per-slice percentile stretch; stored data intact). Black marks voxels where $m \times w(t) > 0.35$, m the heuristic lesion-candidate field: a **prescribed overlay, not predicted growth**, on the axial slice with most mask support (Z = 26/27/38).

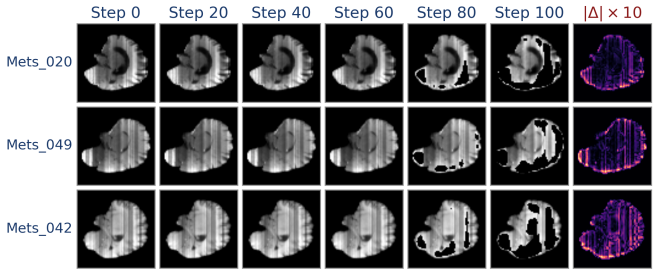


Fig. 3. Three cases, Step 0→100. Black = the **prescribed mask overlay, not predicted growth**. Right: 10× the model's own change (raw intensity; saturates at 0.1).

7. Generation validity verification: sampler surrogates (Fig. 4)

As trained, replacing the sampler with frozen sampling, a constant latent or another patient's result leaves **all four metrics bit-identical**, while source copy and pure noise do (our trivial baselines). This survives the region fix of §5: on the 105 labelled studies all five sampler-side surrogates give **identical** mean change (−0.0494±0.0237), and deleting generation moves it by 16%. Root cause: the decoder's Sigmoid saturates (latent magnitude 11.43; 99.97% of candidate-core voxels exceed the 0.08 clip), so §4–§5 track the post-processing chain, not the sampler.

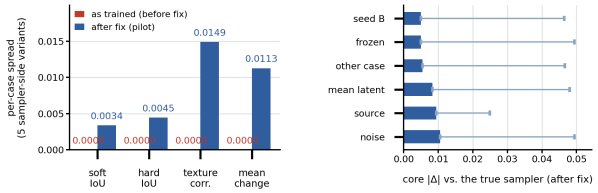


Fig. 4. Sampler attribution before (a) and after (b) the §8 fix.

8. Verified fix: sampler attribution restored

Two changes only (EMA latent standardisation; Hardtanh(0,1) decoder output) under an otherwise identical recipe (4.5 min pilot) give reconstruction PSNR 15.993±1.497 → **29.299±1.318 dB** and decoder std 0.0010 → 0.1447. Re-running the audit on all 51 cases, the five sampler-side variants stop being bit-identical (Fig. 4): the sampler now moves every reported metric, with a post-processed core difference from the true sampler of 0.0050–0.0106. The generation is still not lesion-specific (core/ring 0.94).

9. Take-home and limits

Contribution. A lightweight two-stage latent diffusion transformer that synthesises bounded, reproducible brain-metastasis progression from a single scan, with released code, weights, scripts and checker; expert-lesion calibration of a heuristic mask; and, as a verification layer, a generation-isolation audit plus a two-change decoder fix that restores sampler attribution on all 51 cases. **Limits.** No longitudinal ground truth (the step index is a generation condition, not clinical time); T_d uncalibrated; volumes resampled to 64^3, so spacing is not modelled; the expert-lesion check uses training studies, as labels exist only there.

Compliance with Ethical Standards

Data and ethics. BrainMetShare [5] (156 studies; publisher's split 105/51), non-commercial research licence. Retrospective, de-identified secondary analysis: no new data, no ethics approval or consent required. **Conflicts of interest / funding:** none. **Reproducibility.** Code, weights and scripts released; a paper-to-code checker re-derives every claim (246 pass / 0 fail). **AI disclosure.** OpenAI Codex was used for (i) *code assistance* (training, inference, audit and figure scripts) and (ii) *paper layout and typesetting*; every number is emitted by the released code, and the author directed and edited all content.

References

[1] J. Ho, A. Jain, P. Abbeel, "Denoising Diffusion Probabilistic Models," NeurIPS, 2020.
[2] R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR, 2022.
[3] W. Peebles, S. Xie, "Scalable Diffusion Models with Transformers," ICCV, 2023.
[4] V. Biller et al., "A Biophysically-Conditioned Generative Framework for 3D Brain Tumor MRI Synthesis," arXiv:2510.09365, 2025.
[5] E. Grøvik et al., "BrainMetShare" (data and seg), Stanford AIMI, 2020, DOI 10.71718/z66c-qr59.